

Latency Budget

Per-service targets at p50, p80, p90 and p99 for the authorization path and every value-added service, with composed path budgets by scenario, published industry benchmarks to judge them against, and the ownership rules that make a budget mean something.

Version	1.1
Units	Milliseconds
Scope	Platform-added latency
Companion to	Spec v3.6 §26

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How to read these numbers

Rule	Statement
Unit	Milliseconds. All figures are targets, not observed values
Scope	Platform-added latency only. Issuer, network and cardholder interaction time is excluded and called out in §06
Clock	Starts when the request is accepted at the edge, stops when the network message is handed to the processor connection, plus the return leg. See §09
Budget meaning	A hard deadline enforced by the orchestrator, not an observability threshold. On breach the module's declared disposition applies
Percentiles	Measured per service, per region, over a rolling one-hour window, with a minimum sample floor
Cold start	Included. A budget that only holds on warm paths is not a budget

The arithmetic error to avoid

The p99 of a composed path is **not** the sum of the component p99s. Adding them produces a worst-case figure that assumes every service has a bad tail on the same request, which for largely independent services is vanishingly rare. Sum the p50s to get roughly the composed p50; the composed p99 is materially lower than the summed p99s and is stated separately in §04 as its own target. Anyone reconciling §01 against §04 by addition will conclude the numbers are wrong — they are measuring different things.

Core authorization path

Services every card authorization traverses. Parallel groups are marked; services within a group run concurrently and contribute their maximum, not their sum.

Service	p50	p80	p90	p99	Group	
Edge — authn, idempotency, schema, rate	3.0	5.0	6.5	9.0	Serial	
Resource — payment, session, attempt	1.5	2.2	2.6	3.0	Serial	
Vault — credential reference resolution	0.8	1.2	1.5	2.0	Serial	
Context assembly + config snapshot read	1.2	1.8	2.2	3.0	Serial	
Transaction classification	0.4	0.6	0.8	1.0	Serial	
Screening — sanctions, watchlist	4.0	7.0	9.0	12.0	Parallel A	
Constraint bind — action space	1.0	1.8	2.2	3.0	Parallel A	
Credential selection	0.8	1.5	2.0	3.0	Parallel A	
Flow resolution — plan lookup	0.6	1.0	1.4	2.0	Serial	
Experiment assignment — stable hash	0.3	0.6	1.0	2.0	Serial	
Risk — prediction vector	8.0	14.0	17.0	22.0	Serial	
Authentication decision	1.0	1.8	2.3	3.0	Serial	
Message intelligence — formatting, enrichment	0.9	1.6	2.2	3.0	Serial	
Routing — processor, network, message model	1.2	2.2	3.0	4.0	Serial	
Processor adapter — canonical to network	0.9	1.6	2.2	3.0	Serial	
Response normalisation — inbound leg	0.7	1.2	1.6	2.5	Serial	
State transition	0.5	0.9	1.2	2.0	Serial	
Orchestration overhead — scheduling, deadlines	2.0	3.5	4.5	6.0	Distributed	
Parallel group A — contributes its maximum			p50	p80	p90	p99
max(screening, constraint bind, credential selection)			4.0	7.0	9.0	12.0

Value-added services on the path

Services that run only when the transaction, the merchant entitlement or the flow requires them. Each carries its own budget and is additive to the core path when invoked.

Service	p50	p80	p90	p99	When invoked
Network token cryptogram fetch	6.0	11.0	14.0	20.0	Token credential, per-transaction cryptogram
Token vault reference translation	0.6	1.0	1.3	2.0	Every tokenised transaction
Detokenisation in CDE zone	1.2	2.0	2.6	4.0	Adapter, immediately before network message
PIN block translation (HSM)	2.5	4.5	6.0	9.0	PIN debit, card present
3DS decision — exemption evaluation	1.0	1.8	2.3	3.0	SCA-applicable markets
3DS method call setup	2.0	3.5	4.5	7.0	Platform portion only, before ACS
Delegated authentication verification	4.0	7.0	9.0	13.0	Wallet or issuer-delegated flows
Passkey assertion verification	3.0	5.0	6.5	9.0	Passkey-enabled checkout
Fraud — third-party score call	15.0	28.0	36.0	55.0	Merchant uses an external provider
Device fingerprint resolution	1.5	2.6	3.4	5.0	Device data present
Velocity and behavioural lookup	2.5	4.5	6.0	9.0	Risk strategy requires history
AVS normalisation	0.4	0.7	0.9	1.5	Address present
Debit network eligibility resolution	0.5	0.9	1.2	2.0	US debit — from vault-time attributes
Least-cost routing computation	1.0	1.8	2.4	3.5	US debit, merchant elects LCR
FX rate lock resolution	1.5	2.6	3.4	5.0	Cross-currency, from pushed rate feed
DCC offer construction	2.0	3.5	4.5	7.0	DCC-eligible, before cardholder choice
Attempt strategy decision	1.0	1.8	2.4	3.5	After a non-approval outcome
Surcharge and fee calculation	0.8	1.4	1.8	3.0	Surcharging merchants
Instalment plan resolution	1.2	2.0	2.7	4.0	Instalment products

The one that breaks the budget

A third-party fraud call at 55 ms p99 is larger than the entire rest of the platform path. Where a merchant requires an external provider, the composed target in \$04 does not apply and a separate contracted budget is agreed with that merchant. The alternative — absorbing a vendor's tail into our SLO — makes our number a fiction we do not control.

Resolved ahead of the path

Services that contribute zero to authorization latency because their work happened earlier. Each has an SLO measured in its own context, and the reason it is off the path is stated.

Service	On-path cost	Where the work happens	Its own SLO
BIN and instrument intelligence	0	Vault time, when the credential is stored or a snapshot changes	p99 800 ms at vault time
PINless eligibility resolution	0	Vault time, versioned pure function	Included above
Issuer behaviour profile	0	Precomputed, pushed with the routing table	Refresh p99 5 min
Effective configuration snapshot	0	Compiled on change, pushed and cached in process	Propagation p99 60 s
Constraint set compilation	0	Control plane, offline	Propagation p99 60 s; emergency p99 5 s
Execution plan and Flow compilation	0	At registration, signed bundle	Distribution p99 60 s
Routing and cost tables	0	Pushed, held in memory	Refresh p99 5 min
Risk model artefacts	0	Loaded at deploy, co-located	Load p99 30 s
Sanctions list	0	In-memory, refreshed out of band	Refresh p99 5 min; emergency 5 s
Account updater results	0	Batch and event-driven, written to the credential	Daily cycle

The design rule behind this table

Nothing on the authorization path issues a query whose latency we do not control. Where a module needs data it does not have, the answer is a push channel, not a lookup. This table is the list of things that would otherwise have been remote calls, and it is where most of the platform's latency advantage comes from.

Composed targets by scenario

Platform-added latency for complete paths. These are independently measured targets, not sums of \$01 — see the note in \$00.

Scenario	p50	p80	p90	p99
Card CNP authorization — full path with risk and authentication decision	22	33	41	58
Card CNP — risk disabled by merchant	14	21	27	38
Card CNP — network token with cryptogram fetch	28	42	53	74
Recurring MIT — no authentication decision, warm context	15	22	28	40
Card present — terminal, EMV, online	16	24	30	44
Card present — PIN debit with HSM translation	19	29	36	52
US debit — least-cost routing	18	27	34	48
Cross-currency with rate lock	24	36	45	63
Capture	10	15	19	28
Refund	11	16	21	30
Void or reversal	9	13	17	25
Retry attempt — strategy decision plus re-execution	19	29	36	50
Vault-only call — Forward API, no decisioning	5	8	10	15
Tokenise only — credential store	7	11	14	20
Card CNP with third-party fraud provider	37	61	77	113

The headline commitment is the first row: **platform-added p99 under 60 ms** for a full card-not-present authorization including screening and risk. Every other row is derived from the same components and differs only in which services are invoked.

Industry benchmarks

Published figures from operators, schemes and specifications, against which the targets above can be judged. Component-level data is scarce because operators rarely publish it; where only end-to-end figures exist, that is noted.

Published operator figures

Figure	p50	p99	Source and scope
Adyen ML inference service on the payment path	20	100	Adyen engineering, 2025. Their model-serving endpoint inside the live payment flow. The closest published analogue to our risk budget
Adyen feature serving	single digits	—	Adyen engineering. Feature platform at tens of billions cardinality
Acquirer connection latency, illustrative	127–203	—	Secondary analysis of Adyen routing: the gap between two acquirers for the same corridor. Indicative of routing’s latency leverage, not a scheme figure
Card authorization, general industry guidance	<200	<500	Payment operations practitioner guidance. End-to-end, not platform-added

Scheme and specification limits

Not performance targets — hard limits past which the transaction is taken out of your hands. They bound the outer envelope rather than describing good performance.

Limit	Value	Consequence of breach
Mastercard processor response window	7 s	Network authorises independently under Stand-In Processing. The decision is no longer the issuer’s or ours
Visa 3DS timeout rule	5 s	Visa requires ACS redundancy across physically separate data centres to hold this
Issuer unavailable, no stand-in	—	Response code 91 — issuer or switch inoperative, timeout with no stand-in
Authorization to clearing, domestic	4 days	Visa maximum window; beyond it the authorization is stale
AFD pre-authorization completion	2 hours	Completion or reversal required after an approval response

Our targets against the benchmarks

Our target	Benchmark	Assessment
Risk p50 8 / p99 22	Adyen inference p50 20 / p99 100	Aggressive by a factor of four to five at the tail. See the note below — this is the most consequential gap in the document
Platform-added p99 58	No published equivalent	Nobody publishes platform-added latency separately, which is why it is a useful thing to be able to state and a hard thing to verify against
Composed end-to-end p99, ours plus issuer, roughly 960	Practitioner guidance p99 <500 end-to-end	The guidance is not achievable by anyone once a p99 issuer tail is included. It likely describes normal-condition or issuer-excluded measurement. Treat as directional only
Processor dispatch to response p99 220	Acquirer corridor differences of 70 to 80 ms	Consistent. Routing choice moves this number materially, which is an argument for routing on latency as well as approval
Screening 12 p99	No published figure	Unverifiable externally. Most operators do not have a bank’s screening obligation on the authorization path at all
Every figure in §06 against the 7 s scheme window	Mastercard 7 s	Ample headroom. Our budgets are not constrained by scheme limits; they are constrained by conversion

The risk budget is the number to challenge first

Adyen publishes a p99 of 100 ms for model serving inside their live payment flow. This document targets 22 ms for the equivalent stage. Two readings are possible and they have very different consequences. Either we are counting differently — their figure is a full inference endpoint including feature retrieval, ours assumes co-located inference over features already in the context, which is a real architectural difference and would explain part of the gap. Or our target is simply unrealistic. Risk is the single largest line in the core path budget, so if the true figure is 60 to 100 ms at p99, the composed 58 ms target in §04 does not hold and every scenario figure moves. This should be settled by a spike against a representative model before any of these numbers are committed externally.

Why component benchmarks are thin

Operators publish end-to-end figures because they are favourable and unfalsifiable, and withhold component figures because they reveal architecture. The three useful comparators here came from an engineering blog, a scheme rulebook and a specification — not from anyone's marketing. When our own numbers are measured, the same logic will apply to publishing them.

External and excluded

Time we do not control and do not commit to. Measured and reported, never included in a platform SLO — conflating the two is how competitors publish flattering numbers.

Component	p50	p80	p90	p99	Status
Processor connection round trip	45	80	110	220	Measured, contracted with processor
Network and issuer authorization	180	320	450	900	Measured, not controllable
Issuer stand-in	120	200	280	600	Measured
3DS frictionless — DS and ACS	600	1,200	1,800	3,500	Excluded from platform SLO
3DS challenge	User interaction — not a latency measure				Excluded entirely
Wallet authorization (Apple Pay, Google Pay)	300	600	900	2,000	Excluded
Open banking consent and authentication	User interaction				Excluded entirely
Third-party fraud provider	15	28	36	55	Excluded from the standard composed target
Merchant network transit	20	45	70	180	Excluded

What we report to merchants

Three numbers, always separated: platform-added, external-dependency, and end-to-end. A merchant seeing 260 ms end-to-end should be able to see that 22 ms of it was ours. Reporting only the end-to-end figure makes our contribution unattributable and our improvements invisible.

Asynchronous service SLOs

Off the authorization path entirely. Different units, different consequences of breach.

Service	p50	p90	p99	Unit
Journal event write	8	25	60	ms
Webhook delivery, first attempt	250	900	2,000	ms
Network token provisioning	400	1,200	2,500	ms
Credential vaulting with enrichment	120	400	800	ms
Decision record to learning store	2	8	30	s
Recovery case opened after unknown outcome	1	5	15	s
Reversal advice issued	2	8	30	s
Transaction visible in operations search	3	12	45	s
Merchant reporting freshness	1	5	15	min
Experiment readout refresh	15	30	60	min
Clearing file generation	Per network cut-off			Batch window
Ledger posting after settlement	10	30	60	min

Ownership and breach policy

Service	Budget owner	Disposition on breach
Screening	Screening	HARD_FAIL — transaction fails, no fallback exists
Constraint bind	Control plane	HARD_FAIL unless a signed, unexpired snapshot is held
Risk	Risk	FALLBACK — merchant policy decides: approve, require authentication, or decline
Authentication decision	Authentication	FALLBACK — conservative default from the constraint set
Credential selection	Credential	FALLBACK — default credential per preference
Message intelligence	Message intelligence	DEGRADE — base message proceeds, absence recorded
Routing	Routing	FALLBACK — static table, health-filtered
Processor adapter	Processing	FALLBACK — alternate processor if permitted
Authorization dispatch	Processing	RECOVER — Recovery Supervisor takes ownership
Orchestration overhead	Platform	Escalation — the orchestrator cannot fall back to itself

Rule	Statement
A breach is a defect	A service persistently over budget is a product regression, raised and tracked like any other. It is not an observability metric to be watched
The deadline is enforced, not observed	The orchestrator cuts the call at the budget and applies the disposition. A slow service does not get to consume another service's headroom
Budgets are not negotiable by the optimizer	Latency is a constraint, not a reward term. A strategy cannot trade latency for approval, however favourable the exchange looks
Fallback rate is reported with latency	A service that meets its budget by falling back is not meeting its budget. The two figures are always shown together
Budget changes are governed	Raising a budget is a change with an author, an approver and a recorded rationale. Budgets drift upward by default otherwise
New services must fit	A new service on the path must take its budget from an existing owner or come with a justified increase to the composed target. There is no unallocated headroom

Measurement definitions

Term	Definition
Platform-added latency	Sum of time spent inside platform-owned services on the critical path, excluding time awaiting an external dependency
Clock start	First byte of the request accepted at the edge, after TLS termination
Clock stop, outbound	Network message handed to the processor connection
Clock resume	First byte of the processor response received
Clock stop, final	Response serialised to the client connection
Service latency	Time from request dispatch to response receipt at the orchestrator, including serialisation and transport between services
Critical path	The longest dependency chain through the flow graph, not the sum of all services
Excluded from every figure	Cardholder interaction, 3DS challenge, wallet authorisation, merchant network transit, issuer processing
Window	Rolling one hour, per region, per scenario, with a minimum sample floor below which the percentile is not published
Cold start	Included in all percentiles. Excluding it produces a number that does not describe the service anyone experiences

These are targets, not measurements

Every figure in this document is a design target derived from the architecture, not an observed value from a running system. They are internally consistent and defensible as a budget, and they should be replaced scenario by scenario with measured distributions as services reach production. A target that survives contact with production unchanged was probably not tight enough.